TP1 - Exercise 4: Answers

Part A - Q3. Main idea of each method

Method	Main idea	Works on
StandardScaler	Subtracts the mean and divides by the standard deviation (mean 0, std 1).	Each feature
MinMaxScaler	Rescales each feature to the range [0, 1] using its min and max.	Each feature
MaxAbsScaler	Divides by the maximum absolute value (range [-1, 1]), keeps the zeros.	Each feature
RobustScaler	Subtracts the median and divides by the IQR; less sensitive to outliers.	Each feature
normalize	Divides each row by its norm (L2 norm = 1).	Each sample (row)

Part A - Q4. Binarization

We used the threshold 0. The values greater than the threshold become 1 and the others become 0. A higher threshold gives more zeros, and a lower threshold gives more ones.

Discussion: why does scaling matter?

k-NN and SVM use distances, so a feature with large values dominates the others. Scaling gives all features the same importance. Gradient-based models converge faster and in a more stable way when the features have similar scales. Decision trees do not need scaling, because they only compare a feature with a threshold, and changing the scale of a feature does not change the splits.

Part B - Q1. OrdinalEncoder

The categories_ attribute shows the list of categories of each feature in alphabetical order. The numbers given are only category codes. For example, Europe = 2 and Asia = 1 does not mean that Europe is bigger than Asia.

Part B - Q3. Label encoding and one-hot encoding

Label encoding gives one integer to each class (0, 1, 2, 3), so it creates an order between the classes that does not really exist. One-hot encoding creates one binary column per class (1 if the sample belongs to this class, 0 otherwise), with no order between the classes.

Part B - Q4. Why one-hot for nominal features?

Nominal features (country, browser...) have no natural order. With arbitrary ordinal codes, a model can think that one category is greater than or closer to another one. One-hot encoding avoids this false order.